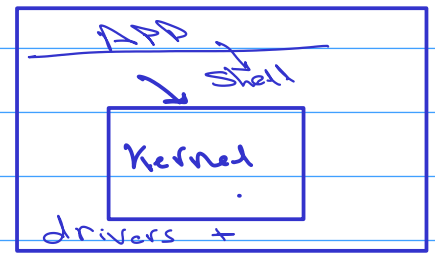


Kernel:-

- Manage processes
- Manage memory
- Command



Shell → Take Commands from user → OS → execute

Key Board

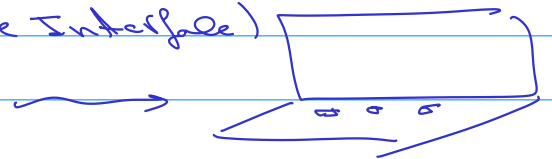


Teletype terminal
TTY



CMD

Shell & terminal (Commandline Interface)



TTY1

Ctrl + Alt + F3

GUI

Terminal working in Background

→ Task Manager

→ g →

TTY2 → Job
↓

GUI →

TTY

← GUI TTY
Residual terminal

TTY1

PTS

Sh

APP

PAS1

Porch

Sh

fork

exit

sh
bash
C Shell
k shell

```

ubuntu 20.04.3 LTS ubuntu tty3
ubuntu_login: itsfoss
Password:
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.11.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

17 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Your Hardware Enablement Stack (HWE) is supported until April 2025.

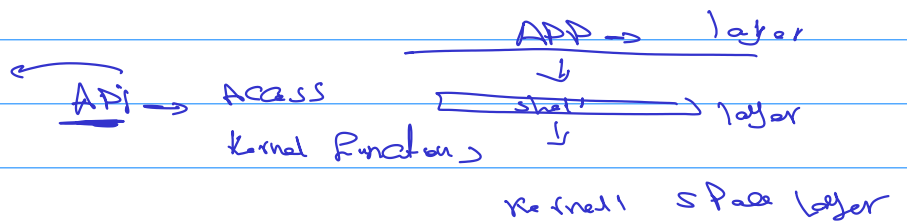
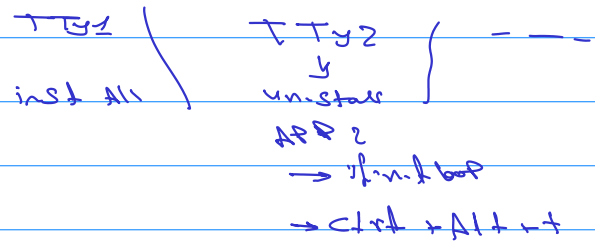
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

itsfoss@ubuntu:~$
  
```

login shell
non-login shell

Main Process



Command → Code → Run → Shared →

create → compile → binary → lib-n → Search → built in command

(1) → stdout

(sorry)

(1) → stdin → input

(2) → stderr → error

after C Application
returning;

Current exit → Int(Signal) ← Ctrl + C

→ infinite →

(n) end

Ctrl + D

Ctrl + L

* Every thing is a Rule *

wer@Machine: ~\$
 Path → CUI
 boot

eslam@eslam:~\$
 eslam@eslam:~\$
 for Infinite
 while(1)
 → Take

~ → home
 (\$) → user
 (#) → root
 ↓
 Privilege

stdin → in

Process → APP

shell ① bin APP
 ②/sbin

usr/local/sbin/usr/local/bin/usr/sbin/usr/bin/sbin/bin/usr/games/usr/local/games/snap/bin/snap/bin/ Built in

(\$) PATH

②

input
 Command :-
 Command, argument, option
 date
 ① variable
 ② int
 ③ file
 Path
 String

Version / APP Path

(.) →

for loop → Print

Path ls -a

Any file begin with

→ hidden

→ APP → bug person →

→ L

→ CC

→ CV

Marketing

1- LinkedIn

2- Open Source → Read

3- Project → Market → Clean → Customize CV

AI → AI → ChatGPT

↑ requirement

↓ knowhow

✓ - help
 ✓ - man

- ChatGPT

long format

```

eslam@eslam:~$ ls -l
total 128380 user g
-rw-rw-r-- 1 eslam eslam 283002 Jan 17 20:23 2026-01-17-Note-20-23.xopp
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Desktop
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Documents
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:47 Downloads
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Music
drwxr-xr-x 3 eslam eslam 4096 Jan 17 20:35 Pictures
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Public
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Templates
drwxr-xr-x 2 eslam eslam 4096 Jan 17 19:18 Videos
-rw----- 1 eslam eslam 135266304 Jan 17 19:38 core.28063
-rw----- 1 eslam eslam 98607104 Jan 17 19:39 core.28507
-rw-rw-r-- 1 eslam eslam 119196224 Jan 17 11:43 google-chrome-stable_current_amd64.deb
drwx----- 9 eslam eslam 4096 Jan 17 21:00 snap
  
```

File → last date → time → File
 drwx → Folder (Directory)
 Size in byte
 ShortCut
 b.nary
 Color Code

rwX, rwX, rwX

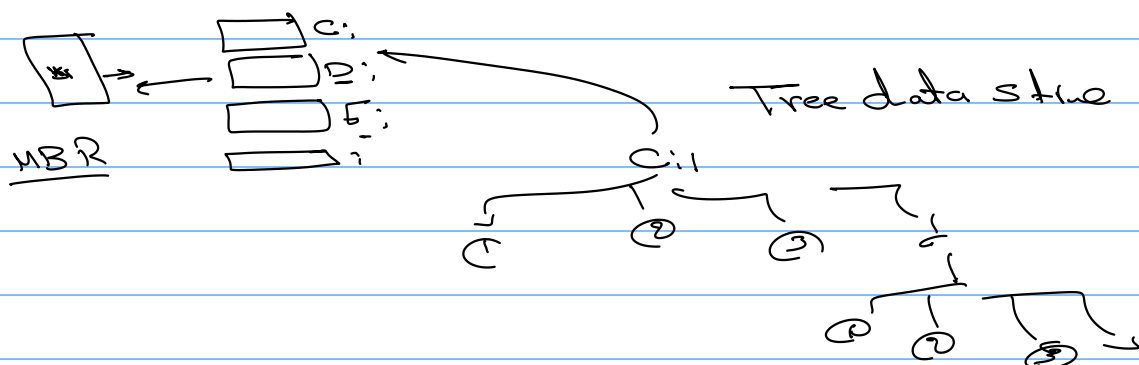
UserManage

Whoami? → C → Print (" ") ;
 Cal → version → 6 → 5
 PWD → Print working directory →
 tar →
 ZIP →
 Kill → Id

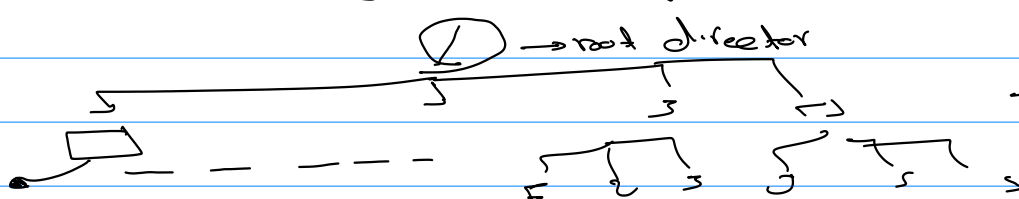
- ① Terminal shell
- ②

File Manager

Window File system

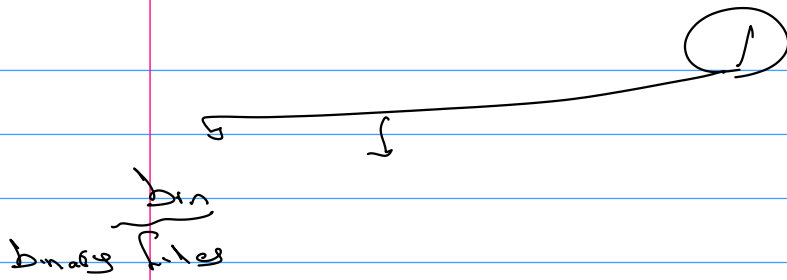


Just one tree



Structure

has special up Pos.



Booting process → on → 
 → 

device → file → 1dev

Editable Text Configuration

etc

/etc/passwd → user
 /etc/shadow →
 /etc/hosts → IP
 /etc/fstab

Code #include <std.lib.h>

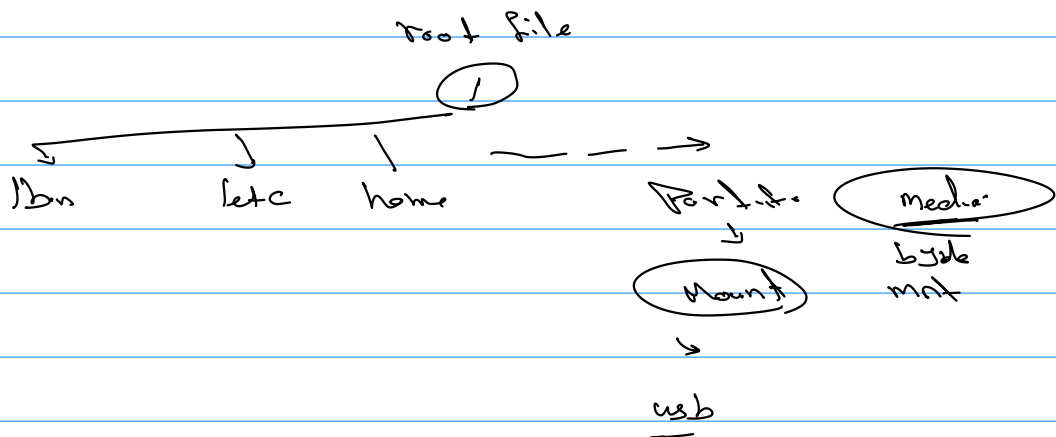
→ static library
 Link → objcc
 ↓
 ← -a /exe./hex

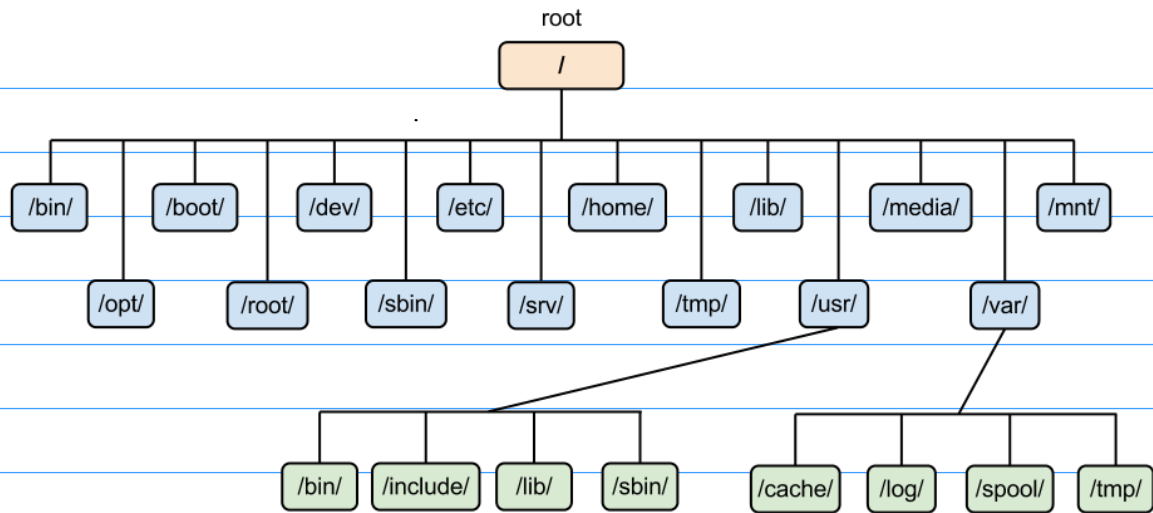
a

dynamic l. brary

Link → reference → memory

← .so Link
 dll windows





io Pins I/O → Force broaden $\equiv$ Limit →

1 - Shell & Command-line interpreter → Command → String

- script as file
- File Manipulation
- printing text
- Program execution

terminal, Program → open window

Prompt: special char that indicates you are ready to type command,

→ `#!/bin/sh`

`/bin/sh`

`/bin/bash`

→ stream →

unix → stream `std out` → file, terminal (1)

`std in` ← file, keyboard (0)

`std err` → file, terminal (2)

- what is command, argument / option

- Task 1. Read file / Try ←

→ Command Try ←

